Software Requirements Specification (SRS)
E-Book Management System

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1. Introduction

The E-Book Management System is a web-based application that enables users to sign up, upload e-books, browse books by categories, and read books uploaded by other users and also download it. The system aims to provide a centralized digital platform for managing electronic books efficiently. It replaces traditional manual book-sharing methods with a secure and accessible digital solution.

1.1 Purpose

The purpose of this document is to provide a detailed description of the functional and non-functional requirements of the E-Book Management System. It serves as a reference for developers, instructors, and evaluators to understand system behavior, scope, and constraints during development and assessment.

1.2 Document Conventions

- FR – Functional Requirement
- NFR – Non-Functional Requirement
- "Shall" indicates a mandatory requirement
- Diagrams follow UML standards
- All terms are defined in the glossary section

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- Software Engineering students
- Project evaluators and faculty
- Developers and testers

Reading order recommendation:

- Introduction
- Overall Description
- System Features
- Non-Functional Requirements
- System Architecture

1.4 Project Scope

The **E-Book Management System** provides:

- User registration and authentication
- Categorized browsing of e-books
- Uploading storing and downloading e-books
- Online book reading functionality

1.5 References

- IEEE 29148 – Software Requirements Specification
- Calibre- E-book Formatting
- Firebase Documentation

2. Overall Description

The system provides a centralized platform where authenticated users can upload and read e-books. It ensures secure access through authentication and stores book files either in cloud storage or local storage as a backup. The system is designed to be simple, modular, and scalable.

2.1 Product Perspective

The E-Book Management System is a standalone web application that operates independently. It interacts with:

- Authentication services
- Database systems

The system follows a client–server architecture.

2.2 Product Functions

Major functions of the system include:

- User Sign Up and Login
- Book Upload and download
- Category-based Book Listing
- Reading E-Books
- Theme Switching (Light/Dark Mode)

2.3 User Classes and Characteristics

User Type	Description
Guest	Can browse categories
Registered User	Can upload, download and read books
Admin	Manages users and categories

2.4 Operating Environment

- Web Browser (Chrome, Edge, Firefox)
- Operating Systems: Windows, Linux, macOS
- Backend Server (Local or Cloud-based)

2.5 Design and Implementation Constraints

- Internet connectivity required for cloud features
- Browser compatibility constraints
- Limited free cloud storage (Firebase) or local database.

2.6 User Documentation

- User guide for signup/login
- Book upload instructions
- Admin management guide

2.7 Assumptions and Dependencies

- Users have basic internet access
- Cloud services are available
- Local database acts as backup

3. System Features (Functional Requirements)

FR-1 User Authentication

- The system shall allow users to sign up and log in securely.
- The system shall validate user credentials.

FR-2 Authorization and Roles

- The system shall enforce role-based access.
- Admins shall manage categories and users.

FR-3 Book Management

- Users shall upload e-books.
- Books shall be categorized and stored.
- Users shall read books online or download them.

FR-4 Data Management

- Users shall create and delete book records.
- The system shall validate uploaded data.

FR-5 Notifications

The system shall display in-app notifications for:

- Successful login
- Successful book upload

FR-6 Activity Logging

The system shall log key user activities such as:

- User login and logout
- Book uploads
- Book access actions

4. External interface requirements

1. User Interfaces

- The system shall provide a web-based graphical user interface (GUI).
- The system shall support responsive design for desktop and mobile browsers.
- Error messages and notifications shall be displayed in a user-friendly manner.

2. Hardware Interfaces

- The system does not require any specialized hardware.
- It shall operate on standard devices including:
  - Desktop computers
  - Laptops
  - Mobile devices
- Storage hardware is abstracted through cloud services or local servers.

3. Software Interfaces

- The system shall interface with the following software components:
  - **Web browser** (Chrome, Firefox, Edge)
  - **Firebase Storage** or **local file system** for storing e-books and authentication
- The frontend shall communicate with the backend using **REST APIs**.

4. Communication Interfaces

- The system shall use **HTTPS** protocol for secure communication.
- Data exchanged between frontend and backend shall use **JSON format**.
- File uploads and downloads shall use secure HTTP methods.
- The system shall support internet-based access for cloud deployment.

5. Non-Functional Requirements

NFR-1 Performance

- Book upload and retrieval operations shall complete within an acceptable response time.

NFR-2 Security

- User passwords shall be securely hashed and never stored in plain text.
- Only authenticated users shall be allowed to upload e-books.

NFR-3 Reliability

- The system shall provide reliable access to users with minimal downtime.

NFR-4 Usability

- The system shall support **light mode and dark mode** for better user experience.
- Users with basic computer knowledge shall be able to use the system without training.

NFR-5 Scalability

- The system architecture shall support an increasing number of users and uploaded books.
- Storage and database components shall be scalable to handle future growth.

NFR-6 Maintainability

- The source code shall follow standard coding practices and documentation.
- The system shall be modular to support easy updates and enhancements.

NFR-7 Portability

- The system shall be deployable on multiple platforms including:
    - Local servers
    - Cloud environments
- The system shall be accessible across modern web browsers and operating systems.

6. System Architecture Overview

1. Presentation Layer

The **Presentation Layer** is responsible for user interaction and interface rendering.

Technologies:

- HTML, CSS, JavaScript
- React (optional)
- Web browser environment

2. Business Logic Layer

The **Business Logic Layer** handles the core functionality and decision-making of the system.

Technologies:

- Node.js / Express.js
- Application controllers and services

3. Data Layer

The **Data Layer** is responsible for storing and managing all persistent data.

4. Integration Layer

The **Integration Layer** enables communication between the system and external services.

7. System and Data Models

Formal UML diagrams, ER models, and workflow diagrams will be provided in supporting design documentation.

8. Validation and Acceptance Criteria

- All requirements shall be tested
- User Acceptance Test (UAT) approval required
- Security and performance testing must pass

9. Appendices

Appendix-A: Glossary

- **E-Book**: Digital book format
- **Authentication**: User identity verification

Appendix-B: Sample Data

- Sample users
- Sample book entries

Appendix-C: Compliance Checklist

- All functional and non-functional requirements are compliant.